

Introduction to Quantum Mechanics

- A) Radiation
- B) Light is made of particles. The need for a quantification
 - 1) Black-body radiation (1860-1901)
 - 2) Atomic Spectroscopy (1888-)
 - 3) Photoelectric Effect (1887-1905)
- C) Wave–particle duality
 - 1) Compton Effect (1923).
 - 2) Electron Diffraction Davisson and Germer (1925).
 - 3) Young's Double Slit Experiment
- D) Louis de Broglie relation for a photon from relativity
- E) A new mathematical tool: Wave functions and operators
- F) Measurable physical quantities and associated operators - Correspondence principle
- G) The Schrödinger Equation (1926)
- H) The Uncertainty principle

Macroscopic world:

A basket of cherries

Many of them (identical)

We can see them and taste others
Taking one has negligible effect
Cherries are both red and good.



Microscopic world:

A single cherry

Either, we look at it without eating, It is red
Or we eat it, it is good.

You can not try both at the same time
The cherry could not be good and red at the same time.



Basics of Quantum Mechanics

Why Quantum Physics?

- **Classical mechanics** (Newton's mechanics) and Maxwell's equations (electromagnetics theory) can explain **macroscopic** phenomena such as motion of billiard balls or rockets.
- **Quantum mechanics** is used to explain **microscopic** phenomena such as photon-atom scattering and flow of the electrons in a semiconductor.
- QUANTUM MECHANICS is a collection of postulates based on a huge number of experimental observations.
- The differences between the classical and quantum mechanics can be understood by examining both
 - The classical point of view
 - The quantum point of view

Classical & Quantum Mechanics

- The motion of the macroscopic particles (heavy particles) can be observed either directly or through the microscope. Classical mechanics can be applied to explain their motion.
- On the other hand, to explain motion of the micro-particles (light particles) like electrons, protons etc. classical mechanics fails.
- Max Planck proposed quantum theory in the beginning of 20th century. In this theory, he proposed that emission of radiations from black body is not a continuous emission but it emits radiations having discrete energy in packets called “quanta of energy” *i.e.* integral value of $h\nu$.
- Einstein latter, applied it to interpret Photo electric Effect *i.e.* the emission of electrons from metal surface

Einstein's Theory

- $E = h\nu$; h = Planck's Constant = 6.63×10^{-34} Js
 - Here light is treated completely as an "object" like a particle. When, it hits a metal, it interacts only with one electron. It may eject the electron or not depending on the energy it has.
 - Incoming energy (photon energy) = energy required to emit the electron + kinetic energy of the emitted electrons.
 - $h\nu = \text{Work Function } (\Phi) + \text{Kinetic energy } (1/2 mv^2)$, where, v = max speed. A frequency vs Kinetic Energy graph should then be a straight line.
- $E_k = h\nu - \text{Work Function } (h\nu_0)$
- Work Function = $h\nu_0$, where ν_0 is "cut off frequency or threshold frequency"- the minimum frequency to achieve the effect.

Applications of Light

First detailed insights into how light interacts with matter-atoms, molecules

-gives understanding to

- ✓ why UV light can cause skin cancer ?
- ✓ why high frequencies are ionizing?
- ✓ It is also the beginning of understanding that
- ✓ why light can cause chemical reactions or changes?
- ✓ why chlorophyll can absorb light?
- ✓ And so on ...

Wave particle duality

Particle:

- ❖ Mass, momentum, position, KE etc.
- ❖ Satisfy Newton's law of motion for mass 'm'
- ❖ Response of motion after collision

$$E = \sqrt{m_0 c^2 + p^2 c^2}$$

Wave:

- ❖ Wavelength, amplitude, frequency phase etc.
- ❖ Wave like motion
- ❖ Response of wave after scattering

$$E = h\nu = hc/\lambda$$

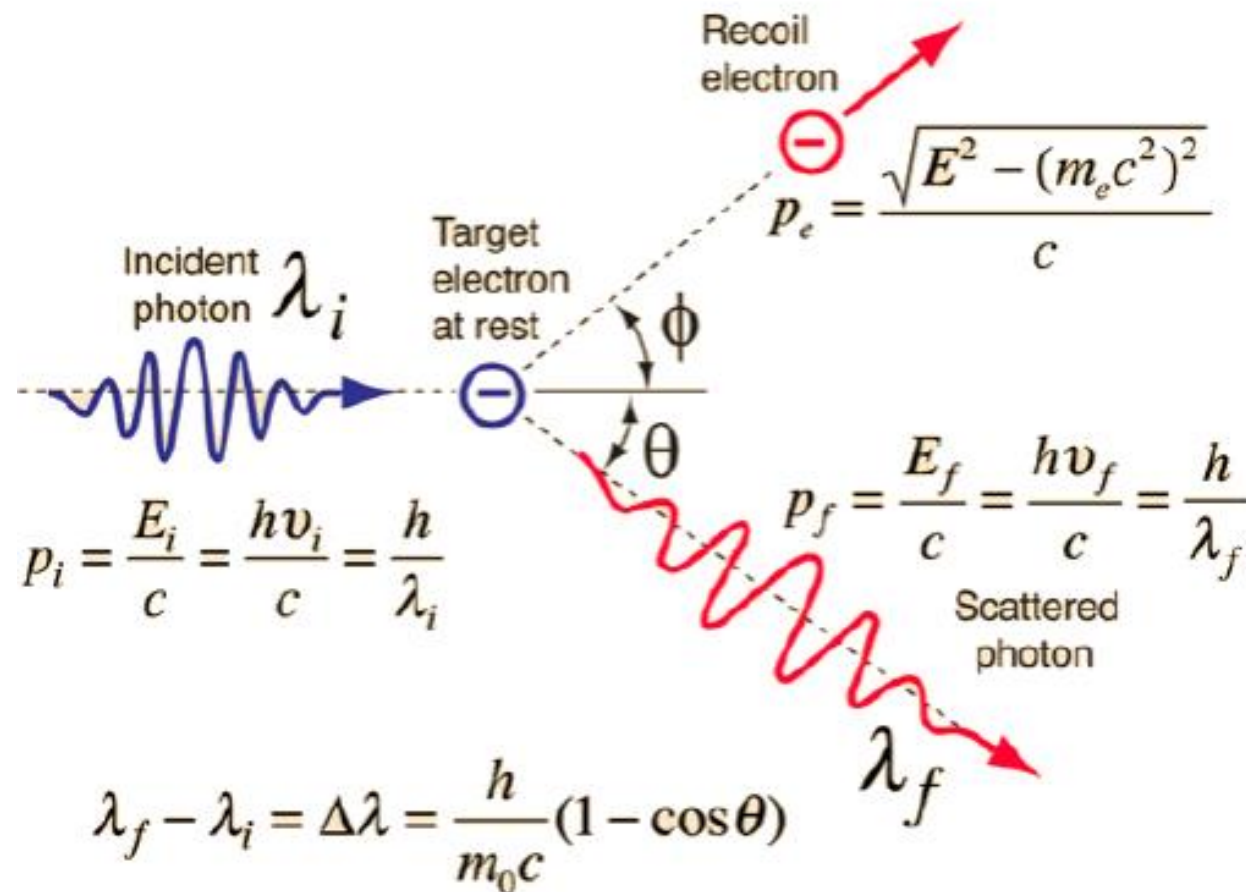
Dual nature of light

An object with definite position in space shows **particle like nature**; where as, an **object with repeatedly periodic pattern specifies wave like motion**.

In physics and chemistry, **wave-particle duality** is the concept that all matter and energy exhibits both wave-like and particle-like properties. A central concept of quantum mechanics, duality, addresses the inadequacy of classical concepts like "particle" and "wave" in fully describing the behavior of small-scale objects. Various interpretations of quantum mechanics attempt to explain this apparent paradox.

Compton Scattering

In graphite, there is an abundance of weakly coupled electrons ---- one can think of them as of “nearly free electrons” or with a good approximation, as of “free electrons”.



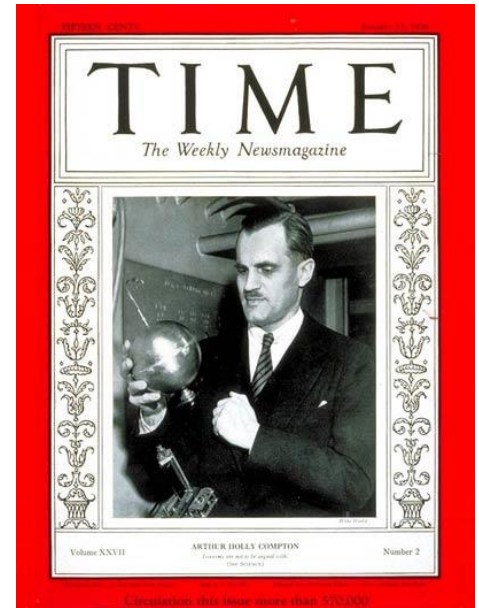
The photon passes some part of its energy to the electron, and flies away with less energy.

The energy has to be Conserved, so the remaining part of the energy is the kinetic energy of the scattered or recoiled electron.

Compton Scattering *playing billiards assuming $\lambda=h/p$*

The phenomenon of scattering of X-ray photon by an electron in an elastic collision is called **Compton effect (1923)**.

- ❑ The scattering is due to inelastic collision of two particles i.e. incident X – ray photon and electron (initially at rest).
- ❑ Conservation of Energy and Momentum has to be applied.
- ❑ The scattered photon has less energy than incident photon



Conservation of Energy

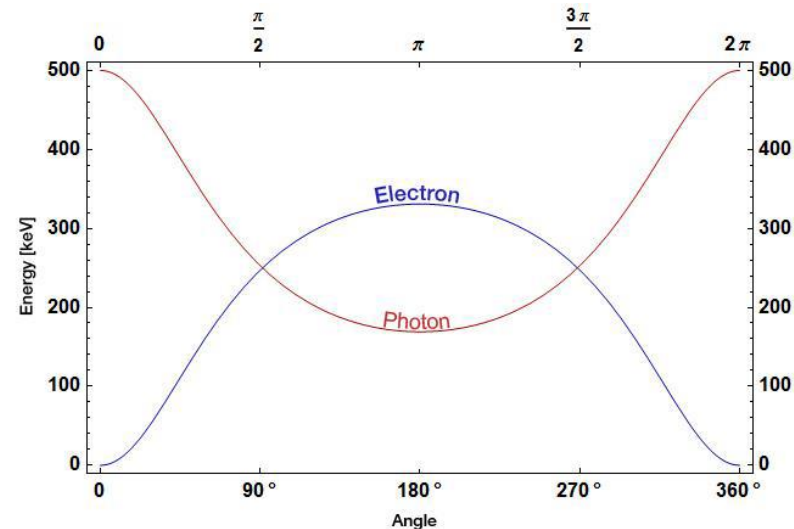
$$h\nu_i + m_0c^2 = h\nu_f + mc^2 \quad \dots(1)$$

where

$$mc^2 = K.E. + m_0c^2 \Rightarrow K.E. = h(\nu_i - \nu_f)$$

Conservation of Linear Momentum

$$p_i = p_f + p_e$$



Conservation of momentum

In the direction of propagation:

$$\frac{h\nu_i}{c} + 0 = \frac{h\nu_f}{c} \cos \theta + p_e \cos \phi$$
$$\text{or } p_e \cos \phi = \frac{h\nu_i}{c} - \frac{h\nu_f}{c} \cos \theta \quad \dots(2)$$

In the direction perpendicular to direction of propagation:

$$0 = \frac{h\nu_f}{c} \sin \theta - p_e \sin \phi$$
$$\text{or } p_e \sin \phi = \frac{h\nu_f}{c} \sin \theta \quad \dots(3)$$

Squaring and then adding Eq. (2) and (3)

$$p_e^2 c^2 = h^2 [\nu_i^2 + \nu_f^2 - 2\nu_i \nu_f \cos \theta] \quad \dots(4)$$

Now squaring Eq. (1)

$$[h(\nu_i - \nu_f) + m_0 c^2]^2 = c^2 p_e^2 + m_0^2 c^4 \quad \dots(5) \quad E = \sqrt{m_0 c^2 + p^2 c^2}$$

Compton Scattering

Now solving Eq. (4) and (5)

$$\lambda_f - \lambda_i = \frac{h}{m_0 c} (1 - \cos \theta) \quad \dots(6)$$

where $\lambda_c = \frac{h}{m_0 c}$

**Known as
Compton wavelength**

$$\Delta\lambda = \lambda_f - \lambda_i = \lambda_c (1 - \cos \theta)$$

This Eq. was derived by A . H. Compton in the 1920 and the phenomenon it describes, which he was the first to observe is known as Compton effect. It constitutes very strong evidence in support of the quantum theory of radiation.

If $\theta = 90^\circ$ then $\cos\theta = 0 \Rightarrow \Delta\lambda = \lambda_c = 0.0242 \text{ \AA}$ for an electron.

If $\theta = 180^\circ$ then $\cos\theta = -1 \Rightarrow \Delta\lambda_{\max} = 0.0484 \text{ \AA}$

which is 0.001% of the visible wavelength, so **Compton scattering can not be observed with visible light.**

$$\Delta\lambda \propto (1 - \cos\theta)$$

Compton Scattering : Relation between Scattering angles

Now again from Eq. (2) and (3)

$$\tan \phi = \frac{v_f \sin \theta}{v_i - v_f \cos \theta} \quad \dots(7)$$

Now from Eq. (6)

$$\frac{1}{v_f} = \frac{1}{v_i} + \frac{h}{m_0 c^2} (1 - \cos \theta)$$

$$\text{or } v_f = \frac{v_i}{1 + 2\alpha \sin^2(\theta/2)} \quad \dots(8)$$

$$\text{where } \alpha = \frac{h v_i}{m_0 c^2}$$

Now from Eq. (7) and (8) $\tan \phi = \frac{v_f \sin \theta}{v_f [1 + 2\alpha \sin^2(\theta/2) - \cos \theta]} = \frac{\cot(\theta/2)}{1 + \alpha}$

$$\text{or } \tan \phi = \frac{\cot(\theta/2)}{1 + \left(\frac{h v_i}{m_0 c^2} \right)}$$

The direction of recoiled electron (ϕ) depends on the direction of scattered photon (θ).

If $\theta = 0^\circ \Rightarrow \phi = 90^\circ$

If $\theta = 180^\circ \Rightarrow \phi = 0^\circ$

**The electron can recoil in forward direction only at $\phi \leq 90^\circ$,
While photon can scattered in any direction.**

Compton Scattering : K.E. of Recoil Electron

Now from Eq. (1)

$$\text{K.E.} = h(\nu_i - \nu_f)$$

$$= h\nu_i \left[1 - \frac{1}{1 + 2\alpha \sin^2(\theta/2)} \right]$$

$$\text{or } \text{K.E.} = h\nu_i \left[\frac{2\alpha \sin^2(\theta/2)}{1 + 2\alpha \sin^2(\theta/2)} \right] \quad \text{where } \alpha = \frac{h\nu_i}{m_0 c^2}$$

$$\text{K.E.} \propto h\nu_i$$

$$\text{If } \theta = 0^\circ \Rightarrow \text{K.E.} = 0$$

$$\text{If } \theta = 180^\circ \Rightarrow \text{K.E. will be maximum}$$

$$\text{K.E.}_{\text{max}} = h\nu_i \left[\frac{2\alpha}{1 + 2\alpha} \right]$$

Compton's Observations

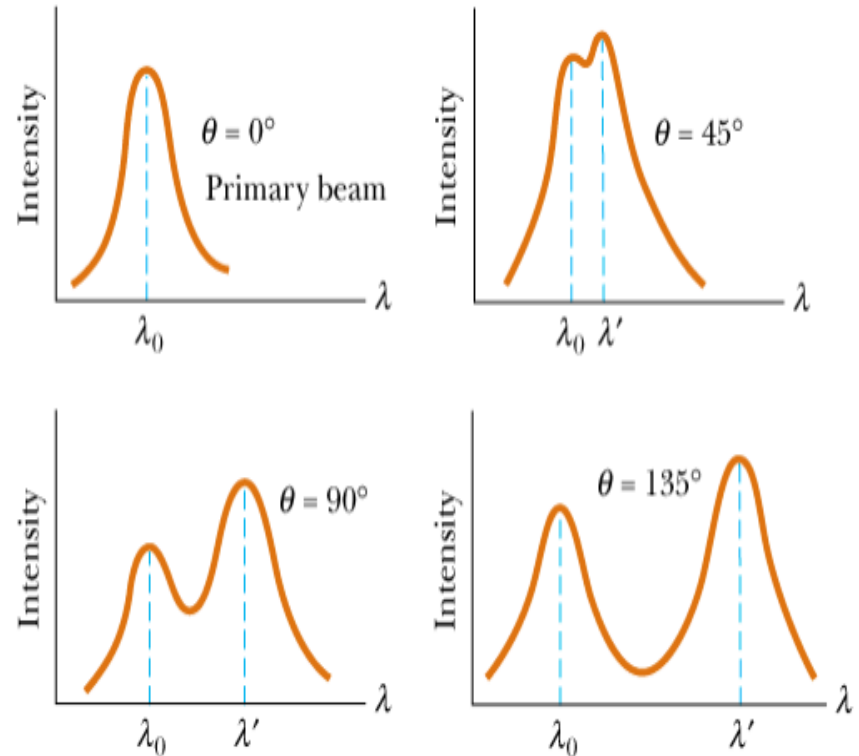
- The scattered radiation had smaller frequency and longer wavelength) than the incident beam.
- The change in wavelength depended on the angle of scattering and not on the material.
- As the collision is elastic, energy and momentum (relativistic) will be conserved
- These could only be explained by quantum theory (existence of quantum particles).

Compton's Observations

For visible light (400nm to 700nm): $\lambda_f - \lambda_i < 0.01\%$ which employs that for visible spectrum, change in wavelength by scattering is very-very small.

For X-rays: $\lambda_f - \lambda_i \approx$ several % for X-ray wavelength ~ 0.1 nm.

This effect is observed clearly for lower wavelengths.



Problems

1. X-rays of wavelength 10.0 pm are scattered from a target. (a) Find the wavelength of the X-rays scattered through 45° . (b) Find the maximum wavelength present in the scattered X-rays. (c) Find the maximum kinetic energy of the recoil electrons.

Ans. 10.7 pm, 14.9 pm, 40.8 keV

2. A photon whose energy equals the rest energy of the electron undergoes a Compton collision with an electron. If the electron moves off at an angle of 40° with the original photon direction, what is the energy of the scattered photon?

Ans. 335 keV